FUNDAMENTAL OF ANIMATION

BRANCH NAME: GAMING AND ANIMATION **SEMESTER – III**

BRANCH CODE: 50

Course Code	Course Title	TH	T	P	TOT	EVALUATION SCHEME						Total Marks	Total Credit
						Internal Assessment		External					
						Theory	Practical	Theory		Practical			
		Periods Per Week				Max Marks	Max Marks	Max Marks	Duration in Hrs.	Max Marks	Duration in Hrs.		
503001	Fundamental of Animation	4	0	0	**4**	30	0	70	2.5	0	-	**100**	**4**
503002	History of Animation	4	0	0	**4**	30	0	70	2.5	0	-	**100**	**4**
503003	2D Animation	4	0	8	**12**	30	20	75	2.5	75	3.0	**200**	**8**
503004	Design Development Advancement	4	0	0	**4**	30	0	70	2.5	0	-	**100**	**4**
503005	Preproduction of Animation and Gaming	4	0	0	**4**	30	0	70	2.5	0	-	**100**	**4**
493001	Data Communication and Networking	4	0	0	**4**	30	0	70	2.5	0	-	**100**	**4**
503007	Photoshop-Fundamentals of Video Editing	4	0	8	**12**	30	20	75	2.5	75	3.0	**200**	**8**
013054	General Proficiency	0	0	4	**4**	0	25	0	-	0	-	**25**	**0**
503052	Industrial Exposure	-	-	-	**0**	0	25	0	-	0	-	**25**	**0**
AS301	Summer Internship (3-4 Week duration After II sem)	-	-	-	**0**	0	25	0	-	25	3.0	**50**	**2**
	Total	**28**	**0**	**20**	**48**	**210**	**115**	**500**	**-**	**175**	**-**	**1000**	**38**

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- Evolution of animation.
- Basic principles of animation.

2. **Traditional Animation Techniques:**

- Frame-by-frame animation: principles & techniques.
- Walk cycles, run cycles & other basic character movements.

3. **Character Animation:**

- Expressions and emotions.
- Lip-sync animation.

4. **Digital Animation Tools and Software:**

- Introduction to popular animation software such as Adobe Animate, Toon Boom Harmony, or Blender.
- Understanding the interface, tools, and workflows of animation software.

5. **Storytelling and Narrative in Animation:**

- Storyboarding.
- Narrative structure, character development, and plot progression in animation.

6. **Team-based Animation Project:**

- Collaborative animation project involving concept development, character design, storyboarding, and animation production.
- Applying animation principles and techniques to create a final animated sequence.

1.0 Introduction to Animation

Animation is a fascinating way of telling stories through moving pictures. It has changed a lot over time, but its main goal remains the same: to make things look like they are moving when they really aren't. How does it do that? Well, imagine you have a bunch of pictures, each showing a slightly different moment in time. When you flip through them very quickly, the pictures blend together, and it seems like the characters and objects in the pictures are actually moving!

There are different types of animation, and each has its own unique charm. Let's talk about a few:

1. Traditional Hand-drawn Animation: This is the classic way of making animations, where skilled artists draw each frame of the animation by hand. It's like making a flipbook with many drawings on each page. When you flip the pages quickly, the characters and scenes come to life.

2. Computer-Generated Imagery (CGI): This type of animation uses powerful computers to create characters and environments in a virtual world. Think of famous animated movies like "Toy Story" or "Frozen." The characters and scenes

are made on the computer, and then they are animated to look like they are moving.

3. Stop-Motion Animation: In this style, physical objects or models are moved and photographed one frame at a time. For example, clay figures or puppets can be manipulated, and pictures are taken after each small movement. When the pictures are played in quick succession, it gives the illusion of motion.

Animation has come a long way since its early days. Nowadays, we can see animated characters in movies, TV shows, video games, and even advertisements! It's a versatile and powerful way of telling stories and entertaining people.

So, next time you watch an animated film or play a video game with animated characters, remember that it's all about bringing still images to life and creating magical worlds where anything is possible!

History and Evolution of Animation:

Early Beginnings: The roots of animation can be traced back to ancient times when people tried to depict movement in their

artworks. For example, cave paintings showed animals in different positions, suggesting motion. Similarly, in ancient Egypt, hieroglyphs were placed in sequence to create a sense of movement.

Inventions in the 19th Century: The real progress in modern animation began in the late 1800s. Two devices, the zoetrope and the praxinoscope, were invented during this time. They used spinning discs with images on them and viewing slits. When you looked through the slits while spinning the disc, the images appeared to move, creating an early form of animation.

Émile Cohl and "Fantasmagorie": In 1908, a French artist named Émile Cohl made a groundbreaking animated film called

"Fantasmagorie."*. This film featured hand-drawn images that were shown in a sequence, making them come to life. It was the first-ever animated film that captured the attention of audiences and inspired other artists to explore animation further.

Advancements in the 20th Century: The 20th century saw significant advancements in animation technology and techniques. Animation studios and filmmakers started experimenting with different styles and storytelling methods. One of the most influential figures in animation history was Walt Disney. He created iconic characters like Mickey Mouse and released the first full-length animated feature film, "Snow White and the Seven Dwarfs," in 1937. This film marked a turning point for animation, as it demonstrated that animated movies could be successful and captivating.

The Rise of Computer Animation: As technology continued to progress, computer-generated imagery (CGI) became a game-changer for animation. In the late 20th century, films like "Toy Story" (1995) showed that entire animated movies could be created using computers, without traditional hand-drawn techniques. This opened up a whole new world of possibilities for animators and filmmakers.

Animation Today: Today, animation is everywhere! We see it in movies, TV shows, video games, advertisements, and even on social media. With the advent of digital tools and software, animators can create incredibly detailed and lifelike animations. They can bring

fantastical worlds, talking animals, and imaginative characters to life with ease.

In conclusion, animation has come a long way from its early beginnings in ancient art forms to the technologically advanced and captivating animations we enjoy today. It has evolved into a powerful form of storytelling that continues to entertain and amaze audiences around the world.

Basic Principles of Animation:

Creating animations that captivate and engage the audience requires understanding some essential principles. Let's explore these principles in easy-to-understand language with examples:

1. Timing:

Timing in animation is like setting the pace of a dance. It determines how fast or slow objects or characters move in the animation. For instance, if a character is taking a leisurely stroll, the timing will be slower, with more frames between each step. On the other hand, if

the character is running, the timing will be faster, with fewer frames between each stride.

2. Spacing:

Spacing in animation refers to the arrangement of movements between key moments. It's like deciding how smoothly or jerkily a character moves. When a character moves naturally, the spacing between frames is consistent, creating smooth motion. In contrast, if a character is being hit by a sudden force, the spacing between frames might be uneven, creating a sense of impact.

3. Squash and Stretch:

Squash and stretch add life and energy to animations. It's like a rubber ball bouncing and deforming when it hits the ground.

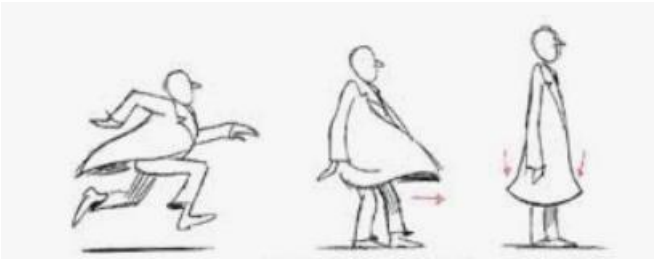
In animation, squash and stretch are used to show weight, impact, and flexibility. For example, when a character jumps, they might stretch upward in the air and squash downward upon landing to give the illusion of weight and impact.

4. Anticipation:

Anticipation is like giving a heads-up before doing something exciting. In animation, it's preparing the audience for an action that's about to happen. For instance, before a character jumps, they might bend their knees slightly and lean backward as a build-up before the actual jump, making the movement more believable.

5. Follow-Through and Overlapping Action:

Follow-through is like the finishing touch after a movement is complete. It's what happens to parts of the character's body after the main action. For instance, when a character stops running, their hair and clothing will continue to move a bit due to their momentum, even after the rest of their body has come to a stop.



6. Exaggeration:

Exaggeration is like adding extra spice to a dish to make it more exciting. In animation, it's used to convey emotions and actions more expressively. For example, if a character is surprised, their eyes might widen dramatically, or if a character is lifting something heavy, their muscles might bulge to emphasize the effort.

By applying these principles, animators can create animations that feel more alive, engaging, and entertaining. Understanding how timing, spacing, squash and stretch, anticipation, follow-through, and

exaggeration work together can elevate the quality of animations, making them more compelling and enjoyable for the audience.

2.0 Traditional Animation Techniques

Traditional animation, also known as hand-drawn animation, is the classic method of creating animations by drawing each frame manually. This technique has a rich history and has been the foundation of animation for decades. Here's a detailed overview of traditional animation techniques:

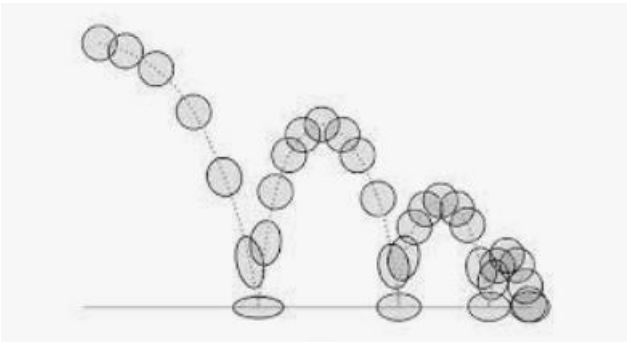
Frame-by-Frame Animation: Principles and Techniques:

Frame-by-frame animation involves creating each frame of the animation individually. Animators draw every single frame, making slight adjustments to the images to create the illusion of movement when played in sequence. This process requires skill, patience, and attention to detail.

1. **Storyboarding:** The first step in traditional animation is creating a storyboard. Storyboarding is a series of rough sketches that outline the key scenes and actions in the animation. It serves as

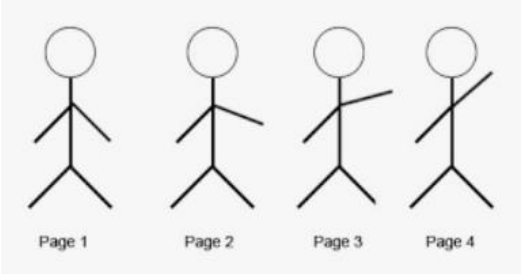
a blueprint for the entire animation process, helping animators plan the sequence and timing of the frames.

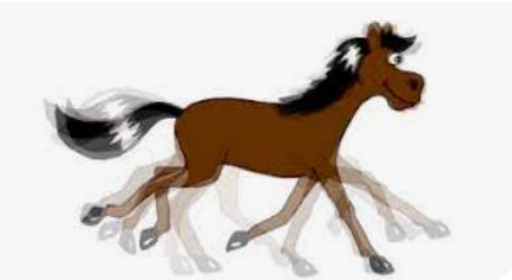
2. **Keyframes:** Keyframes are essential frames that define the major poses or positions of characters or objects during specific points in the animation. They act as reference points for the rest of the animation. For example, in a walk cycle, keyframes would represent the starting pose, the mid-stride position, and the ending pose.



3. **Inbetweening:** After the keyframes are drawn, the animators create in-between frames. Inbetweening is the process of adding frames between the keyframes to make the motion smoother. These frames ensure that the movement appears continuous and natural.

4. **Flipping and Onion Skinning:** To maintain consistency and fluidity, animators use a technique called flipping. They flip through the pages of their animation drawings to see how the motion looks in sequence. Onion skinning, on the other hand, involves viewing multiple frames simultaneously, allowing the animators to see the previous and subsequent frames in a transparent overlay. This helps them maintain consistent proportions and movements.



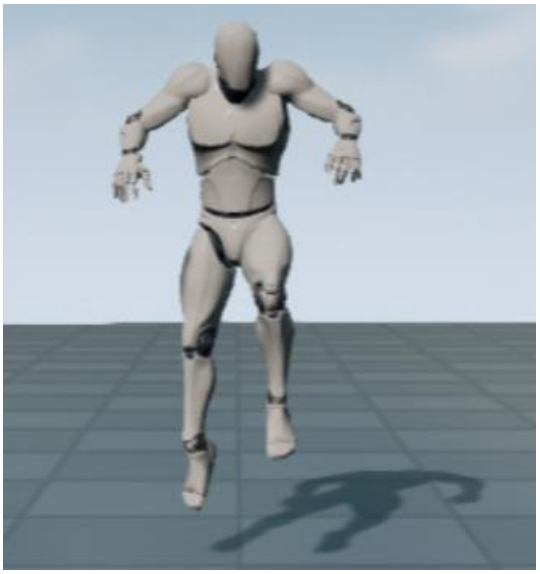


5. **Clean-up:** After the inbetweening process, animators create clean and refined drawings. Clean-up artists trace over the rough animation to remove any unnecessary lines and ensure that each frame is polished and ready for the final animation.

6. **Coloring and Backgrounds:** Once all the frames are completed, the animation is colored, and backgrounds are added.

Backgrounds provide the setting for the animation and are often created separately from the characters to give a sense of depth.

7. **Composite and Finalize:** In traditional animation, the final frames are compiled in the correct order and played to preview the animation. Any necessary adjustments are made at this stage, and additional effects or elements may be added.

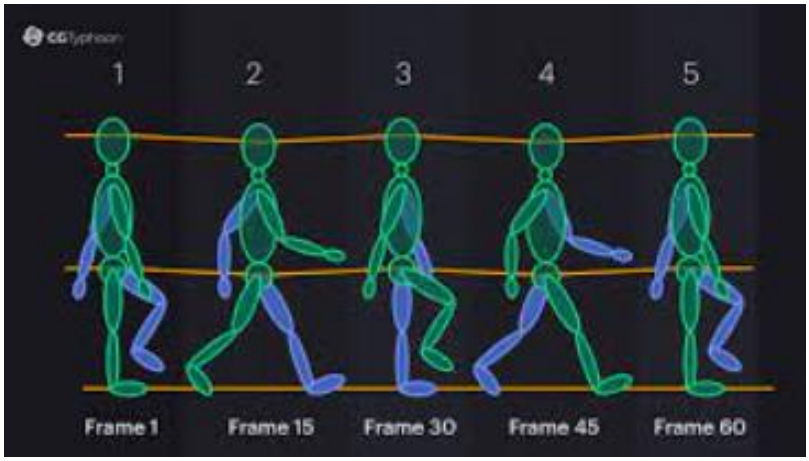


Walk Cycles, Run Cycles, and Other Basic Character Movements:

Walk cycles and run cycles are fundamental animations that form the basis of character movement. Understanding these cycles is essential for animators to create lifelike and convincing character motion. Here's a breakdown of these basic movements:

1. **Walk Cycle:** A walk cycle is a series of frames that depict a character taking steps as they walk. It typically consists of eight key poses:
    - Contact: One foot is on the ground.
    - Down: The foot starts to press into the ground.
    - Passing: The other foot passes the first foot.

- Up: The foot starts to lift off the ground.
- High Point: The foot is at its highest position off the ground.
- Passing: The foot passes the other foot again.
- Down: The foot returns to the ground.
- Contact: Both feet are on the ground.



2. **Run Cycle:** A run cycle is similar to a walk cycle but involves a faster pace. It includes more extreme poses to convey the increased speed and momentum.

3. **Other Basic Character Movements:** In addition to walk and run cycles, animators must master other essential character movements, such as jumping, crouching, waving, and more. Each of these movements requires its own set of keyframes and inbetweening to create believable motion.

3.0 Character Animation

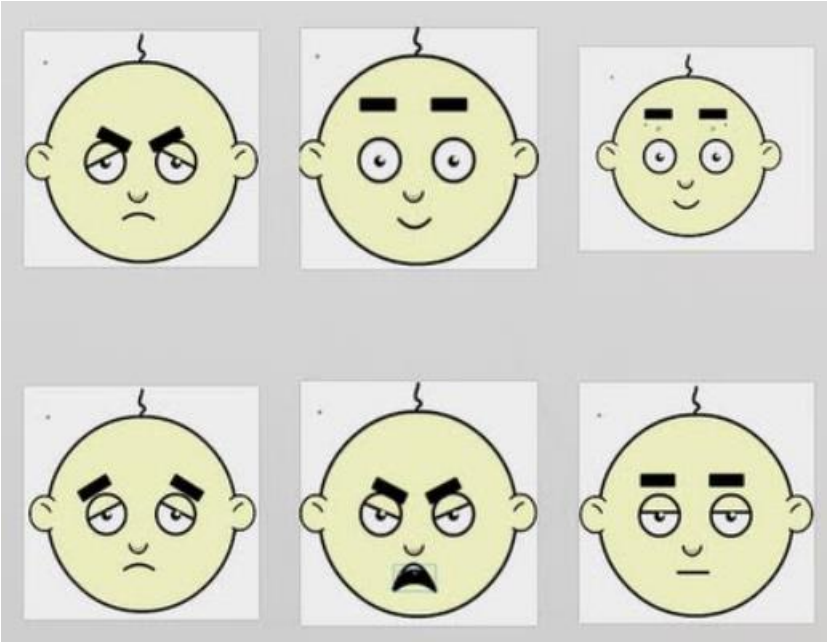
Character animation is a fascinating aspect of animation that focuses on bringing characters to life, imbuing them with emotions, personality, and movements that make them relatable and engaging to the audience. There are two critical aspects of character animation: expressions and emotions, and lip-sync animation.

1. Expressions and Emotions:

The ability to portray emotions effectively is vital in character animation. It involves understanding human expressions, body language, and acting principles to convey feelings and reactions realistically. Here are some key elements and techniques used in animating expressions and emotions:

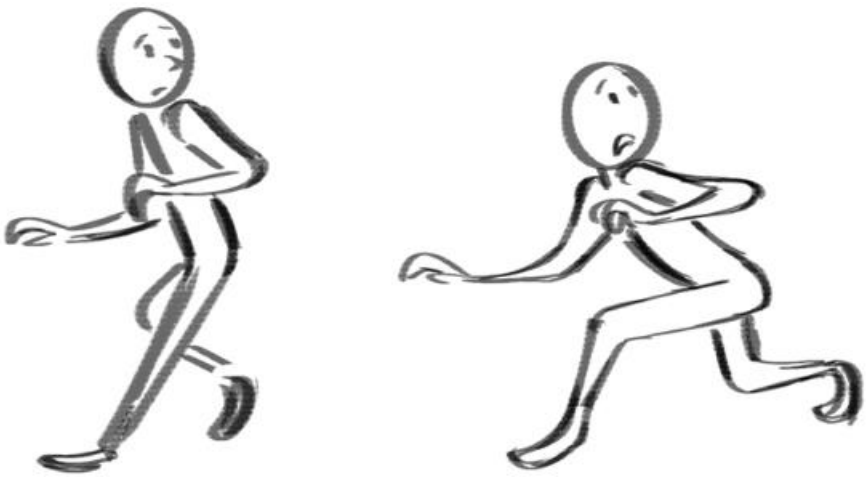
- **Facial Animation:** The face is a powerful tool for expressing emotions. Animators use a character's eyes, eyebrows, mouth, and other facial features to communicate a wide range of feelings, such as happiness, sadness, anger, surprise, fear, and

more. For example, raising the eyebrows can convey surprise, while curving the lips upwards can indicate joy.



- **Body Language:** The body language of a character complements their facial expressions and plays a significant role in conveying emotions. Slumping shoulders may indicate sadness, while clenched fists may show anger or determination.

- **Exaggeration:** Exaggeration is a key principle in animation, and it is especially relevant when portraying emotions. Slight exaggerations can make emotions more apparent and entertaining. However, it's essential to strike a balance to avoid making the animation feel too cartoonish or unrealistic.



- **Reference Videos:** Many animators study reference videos of real people or actors performing specific emotions. This helps them observe natural movements, gestures, and facial expressions, which they can then incorporate into their character animation.

Example: Imagine animating a scene in which a character receives unexpected news. The animator would begin by understanding the

character's personality and the nature of the news. The facial expression might involve widened eyes and raised eyebrows to express surprise, coupled with a dropped jaw to show astonishment. The body language could include the character taking a step back or clasping their hands to their mouth, conveying shock and disbelief.

2. Lip-Sync Animation:

Lip-sync animation involves synchronizing a character's mouth movements with dialogue or sound. This is crucial for creating realistic conversations and ensuring that the characters' lips match the spoken words. Here are some steps and techniques used in lip-sync animation:

- **Phonemes:** Animators use phonemes as the basis for lip-sync animation. Phonemes are the distinct sounds that make up speech. Different sounds require specific mouth shapes to accurately represent them.
- **Audio Analysis:** Animators analyze the dialogue or sound to identify the phonemes and the timing of each spoken word.

They divide the dialogue into individual sounds and determine the appropriate mouth shapes for each.

- **Mouth Charts:** Mouth charts are references that display the different mouth shapes or phonemes. These charts help animators maintain consistency in the character's lip-sync throughout the scene.

- **Pose-to-Pose Animation:** Animators animate the key mouth shapes first and then create in-between frames to ensure smooth transitions between the phonemes. This process is similar to traditional animation techniques, ensuring that the

mouth movements flow naturally and accurately depict the speech.

Example: Suppose a character in an animated movie is saying the word "Hello." The animator would identify the phonemes "H," "eh," "l," and "oh" and animate the character's mouth movements to form these shapes in sync with the audio. The mouth might close for the "H" sound, open slightly for the "eh" sound, form an "L" shape for the "l," and then round for the "oh" sound.

In conclusion, character animation involves an intricate blend of art, observation, and technical skills. Animators use expressions, emotions, and lip-sync animation to breathe life into characters and create captivating stories that resonate with the audience. By understanding human behavior, studying references, and mastering the principles of animation, animators can create characters that evoke genuine emotions and immerse the audience in their animated worlds.

4.0 Digital Animation Tools and Software

Digital animation software has revolutionized the animation industry, offering a wide range of powerful tools and features that streamline the animation process and allow animators to bring their creative visions to life more efficiently. Here, we'll provide an introduction to some popular animation software, namely Adobe Animate, Toon Boom Harmony, and Blender, along with an overview of their interfaces, tools, and workflows.

1. Adobe Animate:

Adobe Animate, formerly known as Adobe Flash, is a versatile animation software widely used for creating 2D animations, interactive multimedia content, web animations, and more. It offers a user-friendly interface and integrates well with other Adobe Creative Cloud applications.

Interface and Tools:

- Timeline: The timeline in Adobe Animate allows animators to manage and organize layers, frames, and keyframes. It provides a visual representation of the animation sequence.
- Drawing and Painting Tools: Adobe Animate provides a range of drawing and painting tools to create artwork directly within the software. It includes brush tools, vector drawing tools, and shape tools for creating characters and backgrounds.
- Symbol Library: The software has a symbol library that allows users to create reusable elements like characters, objects, or animations, which can be easily dragged and dropped into the canvas.

Workflow Example: An animator using Adobe Animate can create a simple character animation by drawing each frame of the character's movement on different layers within the timeline. They can then use the symbol library to create reusable symbols for body parts like arms, legs, and facial expressions. These symbols can be manipulated and animated to create seamless movements.

2. Toon Boom Harmony:

Toon Boom Harmony is a professional-grade animation software used for both 2D and 3D animations. It is favored by many animation studios and is known for its powerful tools and robust features.

Interface and Tools:

- Node View: Toon Boom Harmony features a Node View, which allows animators to create complex and interconnected effects, composites, and transformations. It provides a high level of control over the animation elements.
- Bone Rigging: The software offers advanced bone rigging tools that enable animators to create complex character rigs, making character animation more efficient and flexible.
- Effects and Compositing: Toon Boom Harmony provides a wide range of effects and compositing tools to enhance the visual appeal of animations. Animators can add shadows, lighting, and particle effects to their scenes.

Workflow Example: Using Toon Boom Harmony, an animator can create a character rig by placing bones in the character's body parts. They can then animate the character using the bone rig, achieving smoother and more natural movements. The Node View can be utilized to add effects and compositing elements, creating a visually appealing final animation.

3. Blender:

Blender is an open-source 3D animation software known for its extensive capabilities, including 3D modeling, rendering, and animation. It is favored by animators and artists for its flexibility and the fact that it is free to use.

Interface and Tools:

- 3D Viewport: Blender's interface revolves around a 3D viewport, where users can create, manipulate, and animate 3D objects and characters.

- Armature and Rigging: Blender offers a robust armature system for rigging 3D models, allowing animators to create intricate character animations with realistic movements.
- Keyframe Animation: Animators can set keyframes in Blender to define an object's position, rotation, and scale at specific points in time, enabling smooth and controlled animation.

Workflow Example: In Blender, an animator can model a 3D character, apply an armature for rigging, and then use keyframe animation to create movements. They can pose the character by adjusting the bones and setting keyframes at different points in the timeline to achieve the desired animation.

In conclusion, these popular animation software options offer unique features and workflows suited to different animation needs. From 2D character animation to complex 3D projects, each software provides a range of tools to cater to the diverse requirements of animators and artists. As animators explore and become proficient in these

software packages, they can unleash their creativity and produce captivating animations for various media platforms.

5.0 Storytelling and Narrative in Animation

Animation is not just about creating moving images; it's about telling compelling stories that resonate with the audience. Storytelling in animation involves crafting engaging narratives, well-developed characters, and coherent plot progressions. Two essential elements of storytelling in animation are storyboard and narrative structure.

1. Storyboarding: Creating Visual Narratives and Planning Shots:

Storyboarding is the process of visualizing the narrative through a series of illustrated frames. It serves as a blueprint for the animation, allowing the animators and directors to plan the shots, camera angles, and overall flow of the story. Storyboards consist of rough sketches that capture the key moments and actions in the animation.

Importance of Storyboarding:

- Visualization: Storyboards help translate the script or screenplay into a visual form, allowing the team to see how the animation will unfold on screen.

- Pacing and Timing: Storyboards help determine the pacing and timing of scenes, ensuring that the animation flows smoothly and captures the right emotions and impact.
- Collaboration: Storyboards facilitate effective communication among the creative team, including animators, directors, and designers, by providing a clear visual reference.

Example: Consider an animated short film about a young adventurer searching for a lost treasure. The storyboard will outline the protagonist's journey, key locations, challenges, and pivotal moments. Each storyboard frame will depict important scenes, character actions, and camera angles, providing a visual guide for the animation process.

2. Narrative Structure, Character Development, and Plot Progression in Animation:

A well-structured narrative is crucial for captivating the audience and holding their attention. In animation, just like in any other storytelling medium, a narrative typically follows a specific structure

that includes a beginning, middle, and end. Additionally, character development and plot progression play essential roles in creating a compelling story.

Narrative Structure:

- **Introduction/Exposition:** The beginning of the narrative introduces the setting, main characters, and the central conflict. It establishes the context and provides essential background information.
- **Rising Action:** The rising action comprises a series of events that build tension and propel the story forward. It includes challenges, obstacles, and character growth.
- **Climax:** The climax is the turning point in the story, where the central conflict reaches its peak. It is the most intense and emotionally charged moment in the animation.
- **Falling Action:** After the climax, the falling action shows the consequences of the characters' actions and the resolution of subplots.

- **Conclusion/Resolution:** The conclusion brings the story to a satisfying end, resolving the central conflict and providing closure for the audience.

Character Development:

- Characters should have clear goals, motivations, and personalities that drive their actions throughout the animation.
- They should undergo growth and change over the course of the story, facing challenges that test their resolve and character traits.

Plot Progression:

- The plot should unfold in a logical and coherent manner, with each event leading naturally to the next.
- Well-timed reveals and plot twists can add intrigue and surprise to the animation.

Example: In a short animated film about a timid caterpillar who dreams of becoming a butterfly, the narrative structure will introduce

the caterpillar's desires and challenges at the beginning. The rising action will show the caterpillar's efforts to overcome obstacles, while the climax will be the moment of transformation into a butterfly. The falling action will depict the butterfly's struggles and triumphs in its new form, leading to the conclusion where it fulfills its dream of flying freely.

In conclusion, storytelling in animation involves effective use of storyboarding, narrative structure, character development, and plot progression to create engaging and memorable animations. By combining these elements thoughtfully, animators can captivate their audiences and leave a lasting impact with their animated stories.

Team-based Animation Project:

Collaborative animation projects involve a team of talented individuals working together to create an animated sequence. From concept development to animation production, each stage requires careful planning, creativity, and effective communication among

team members. Let's delve into each aspect of a team-based animation project with examples:

1. Concept Development:

The concept development phase is where the team brainstorms ideas, develops the storyline, and defines the overall vision for the animation. This stage includes determining the target audience, setting the tone, and establishing the main themes and messages.

Example: Suppose the team wants to create an animated short film about a young astronaut exploring an alien planet. The concept development phase involves deciding the protagonist's personality, the challenges they will face on the alien planet, and the underlying themes of exploration and discovery.

2. Character Design:

Character design involves creating visually appealing and distinct characters that fit the story's narrative and setting. The team collaborates to develop the main characters, secondary characters, and any supporting creatures or entities.

Example: For the astronaut animation project, the team designs the young astronaut with a futuristic spacesuit, reflecting their adventurous spirit. They also design the alien creatures, considering their appearances, behaviors, and how they interact with the astronaut.

3. Storyboarding:

Storyboarding is a critical step in visualizing the animation's flow and composition. It provides a visual outline of the animation, showing the sequence of shots and key moments.

Example: The team creates a storyboard sequence showcasing the astronaut's journey on the alien planet. The storyboard includes sketches of each scene, depicting the protagonist's actions, the alien creatures they encounter, and the challenges they overcome.

4. Animation Production:

The animation production phase involves bringing the characters and scenes to life. Each team member plays a crucial role in this stage, including animators, background artists, sound designers, and more.

Example: The animators take the storyboard frames and use animation principles (e.g., timing, spacing, squash and stretch) to create fluid and believable character movements. The background artists design the alien landscapes, while the sound designers add sound effects and create an immersive audio experience.

Applying Animation Principles and Techniques:

Throughout the animation process, the team applies various animation principles and techniques to enhance the animation's quality and appeal.

Example:

- For the astronaut's movement, the animators apply the principles of anticipation and follow-through to make their actions feel more natural and realistic.
- When animating the alien creatures, exaggeration is used to make their unique features stand out, adding an element of interest and charm.

Final Animated Sequence:

Through the collective efforts of the team, the animated sequence gradually takes shape. As each element comes together, from the character animation to the background art and sound design, the story unfolds and captivates the audience.

Example: The final animated sequence showcases the young astronaut exploring the alien planet, encountering various challenges and forming a connection with one of the friendly alien creatures. The animation conveys themes of curiosity, friendship, and the wonders of exploration.

In conclusion, a team-based animation project requires effective collaboration, creativity, and the application of animation principles. By combining concept development, character design, storyboarding, and animation production, the team brings the animated sequence to life, creating a captivating and engaging storytelling experience for the audience.